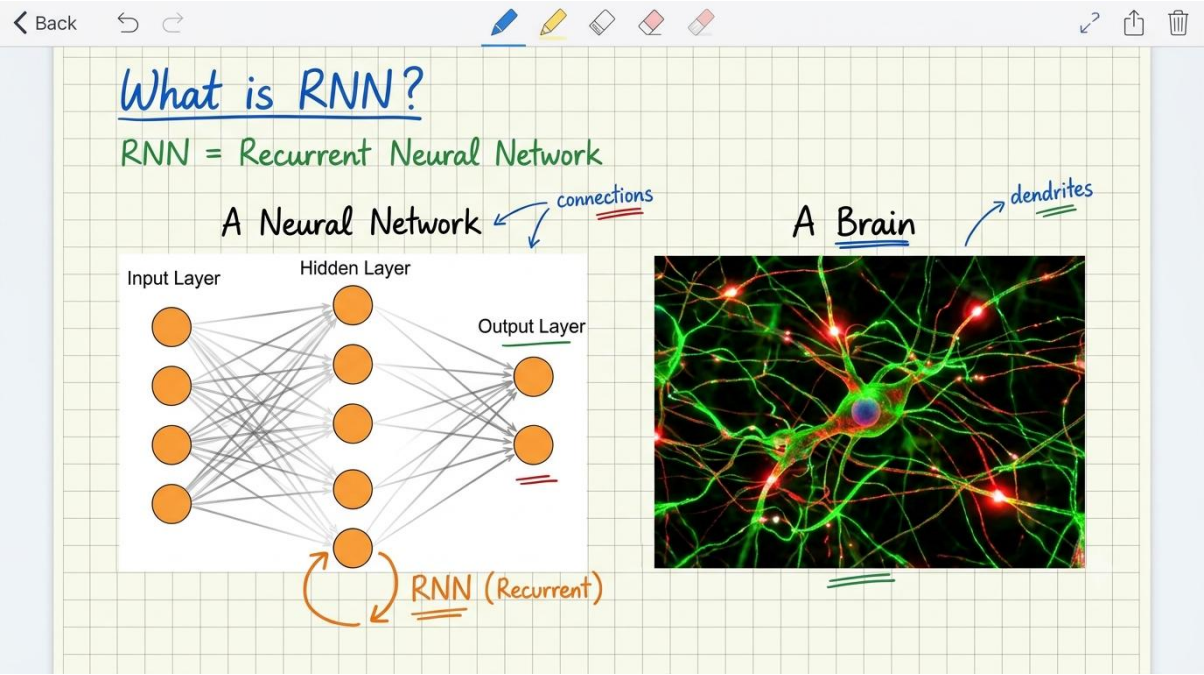
WHAT IS RNN:-



It's a type of neural network used to process sequences, like:

- Sentences (words)
- Time-series data (stock prices)
- Speech/audio signals
- Anything where order (sequence) matters

Why "Recurrent"?

Because RNNs remember previous inputs using a loop. They pass information from one time step to the next.

Imagine:

Input: ["I", "love", "mangoes"]

Time Step 1: Input = "I", Output = h1
Time Step 2: Input = "love", + h1 → Output = h2
Time Step 3: Input = "mangoes", + h2 → Output = h3

So at each step, it:

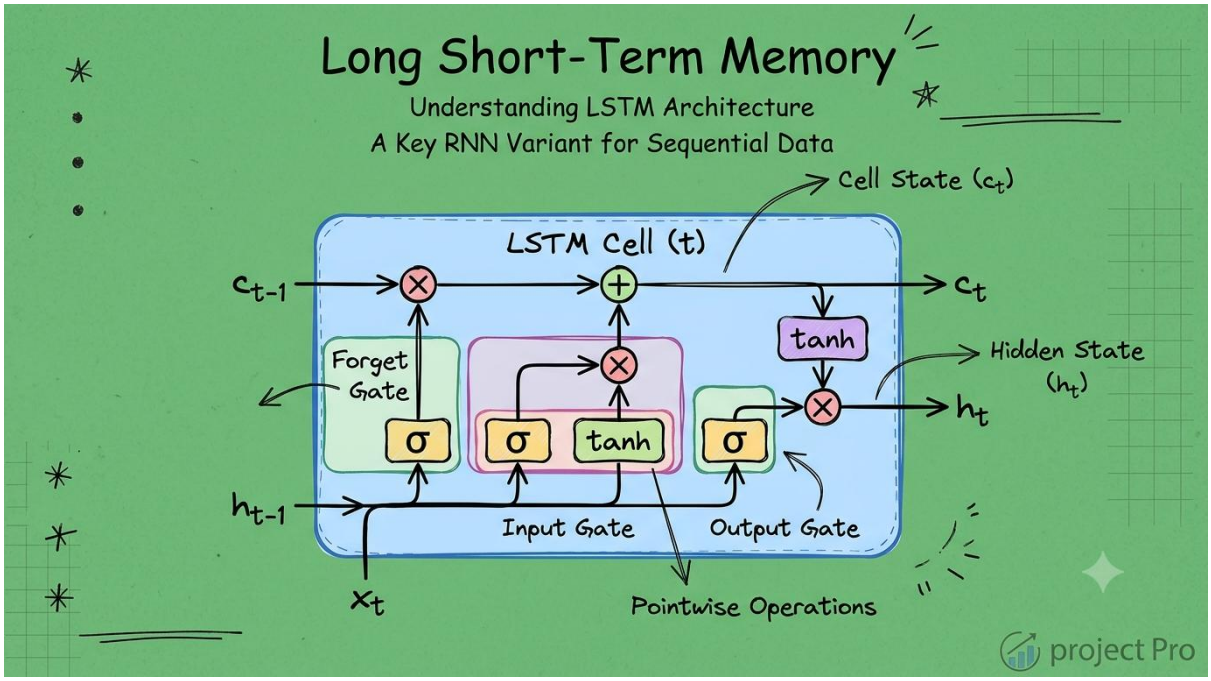
- Takes current input word
- Takes hidden state from previous step
- Produces new hidden state (memory) + output

LSTM = Long Short-Term Memory

LSTM is a special type of RNN (Recurrent Neural Network) that is designed for handling sequence data, but it's smarter and more effective than basic RNNs.

Why Was LSTM Invented?

- Basic RNNs struggle to remember long-term information — they tend to forget things as the sequence gets longer.
- LSTM solves this problem by using memory cells and gates that allow it to remember important information for a longer time and discard what's not useful.



https://colah.github.io/posts/2015-08-Understanding-LSTMs/

Gate	Purpose

Forget Gate	Kya bhoolna hai
Input Gate	Kya naya yaad karna hai
Output Gate	Kya dikhana hai as output

RNN vs LSTM

Feature	**RNN**	**LSTM**
Memory	Short	Long (better)
Handles long-term context?	No	Yes (somewhat)
Structure	Simple	Complex (gates + memory)
Use today?	Rare	Replaced by Transformer

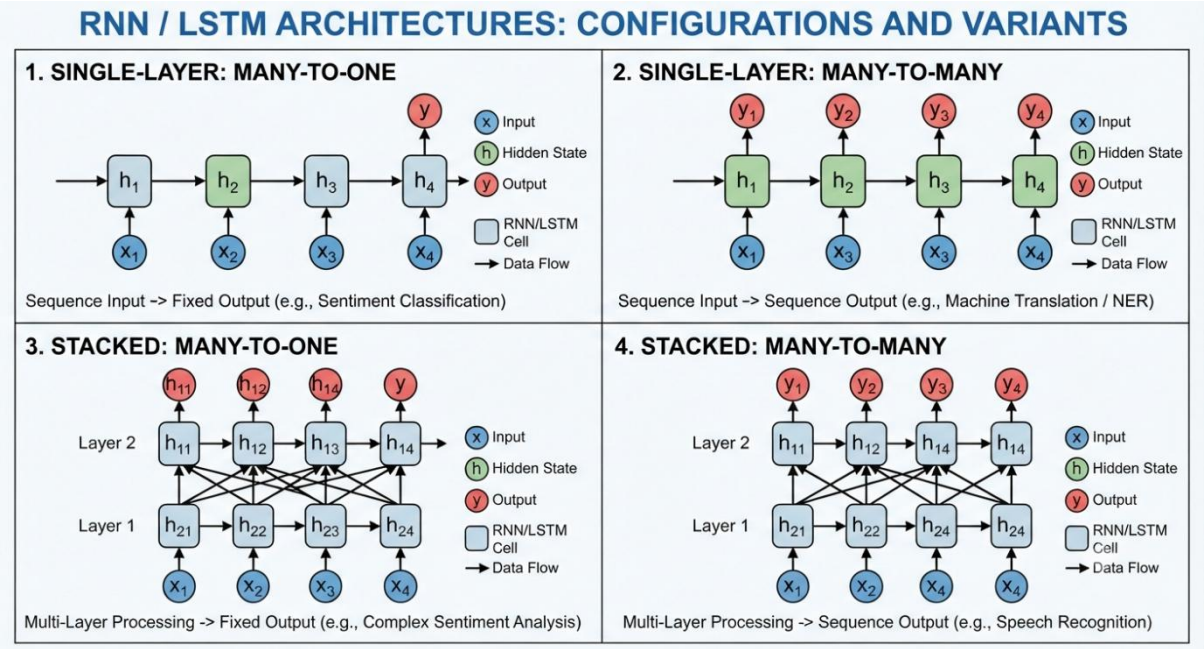
Limitations of Basic RNN

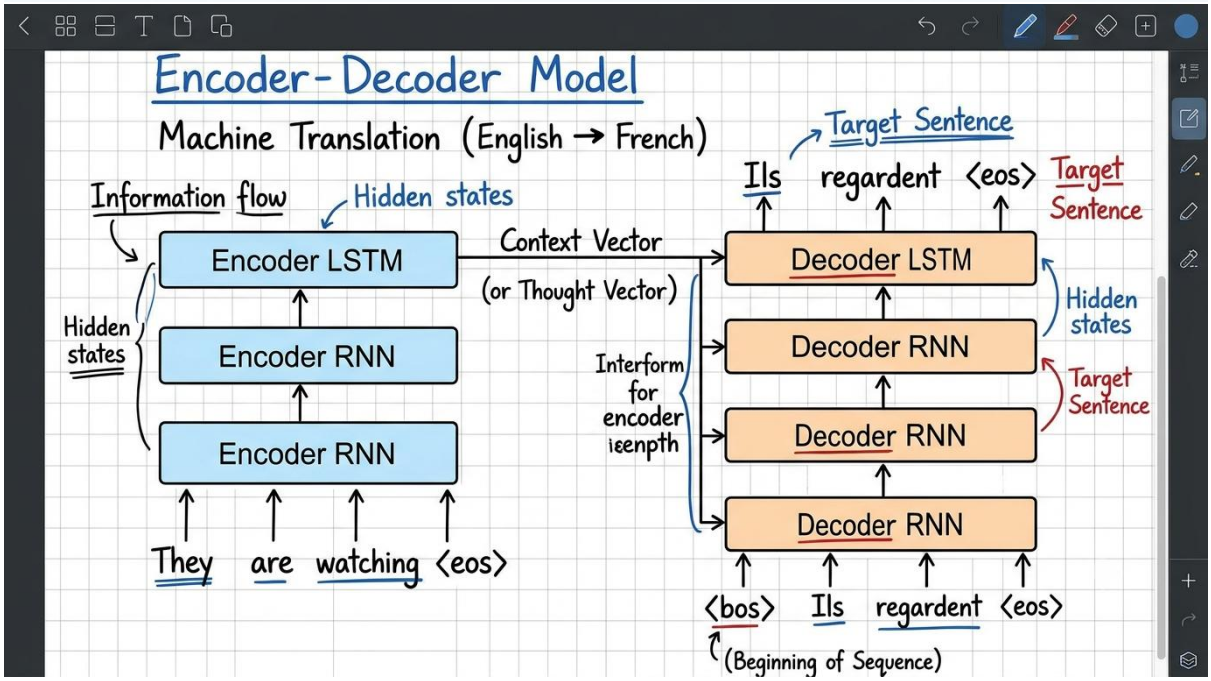
Problem	**Why It Happens**
❌ Forgets long-term info	Because it passes memory step-by-step
❌ Vanishing gradients	During backpropagation over long sequences
❌ Slow training	No parallelization (it's sequential)

What is Sequence-to-Sequence Learning?

Sequence-to-Sequence (Seq2Seq) learning is a neural network approach where:

- Input = a sequence
- Output = another sequence
- Input and output sequences can be of different lengths





https://colab.research.google.com/drive/1E-HI7YKLnPqXJHTbU3ATQU6yzUUogEAu#scrollTo=mNBF_8sZzotG